

Flood Risk Classifier

Model #1 of the AquaGuard / Bangladesh Flood Early-Warning System

A temporal, live-weather-driven flood/no-flood forecaster — literature review,
methodology, real data, and evaluation results

August 14, 2026

Companion to the discharge-forecaster and flood-susceptibility reports

1. Introduction & Purpose

This model answers the question the whole early-warning system exists for: **will this station flood in the next 24, 48, or 72 hours**, using live rainfall, soil-moisture, and river-discharge data. It is deliberately a classifier (flood / no-flood probability), not a single opaque risk score — three independent per-horizon models, each producing its own calibrated probability, matching how real early-warning systems present multi-lead-time risk.

This report documents the model as it exists today, including a 2026-08-14 improvement pass that added probability calibration and a naive-baseline comparison the model never had before — both are covered in full, including a finding that is not flattering but is real (§8).

2. Literature Review

A dedicated, fresh multi-search literature pass (~28 queries) was run against this model specifically on 2026-08-14 — deliberately treated as if the model didn't exist yet, to get unbiased findings rather than re-reading earlier design reasoning. The sources below are the ones that either confirmed an existing design choice independently, or surfaced a concrete improvement.

2.1 Methodology & data requirements

Source	Key finding relevant to this project
Flood forecasting with machine learning models in an operational framework (<i>HESS</i> , Google Research)	Real operational systems split ML use across distinct subsystems (data validation, stage forecasting, inundation, alerting) — this project's own scope (forecasting only) matches the "stage forecasting" subsystem specifically.
Flood early warning system data requirements (multiple sources: NASA Lifelines, IMERG, soil-moisture literature)	Rainfall, river discharge, and soil moisture are the three consistently-cited core inputs — exactly this model's three live feature groups.
Rare event prediction / class imbalance in flash-flood prediction (<i>arXiv</i> , <i>ACM Computing Surveys</i>)	Flash floods occur on "perhaps 1-10% of days" at a given location — a model optimizing plain accuracy would just always predict "no flood" and still score well while being useless. Precision/recall-based evaluation (not accuracy) is essential — matches this model's own PR-AUC/recall-target design.

2.2 Model family choice

Source	Key finding relevant to this project
Rare-event / imbalanced flood prediction (same search as above)	"GBDT are particularly well-suited for imbalanced datasets of rare flooding events, due to gradient boosting on difficult-to-classify examples" — directly supports the LightGBM choice.
A systematic review of flood prediction 2018-2025 (<i>ScienceDirect</i>)	"Clear trend toward ML/ensemble methods... RF, SVR, XGBoost dominate due to interpretability and reliability" alongside deep learning (LSTM/GRU/CNN/Transformer) approaches.
Knowledge-Guided ML for Operational Flood Forecasting (<i>Water Resources Research</i> 2025, Microsoft Research)	A genuinely different SOTA direction: a Factorized Hierarchical Neural Network with separate inverse (infer catchment state) and forward (predict from that state) components — Google/Microsoft-scale infrastructure, judged out of scope for a 30-station project.
Google Flood Hub / Hydrologic Model (<i>Google Research</i>)	ME-LSTM with per-weather-product embedding networks and a CMAL probabilistic output — the reference-class deep learning approach, trained on thousands of global gauges (this project has 30).

2.3 Evaluation methodology

Source	Key finding relevant to this project
Classification approach for flood early warning systems (multiple sources)	"Accuracy alone is insufficient... a model could miss every flood while still maintaining high accuracy" — precision/recall and PR-AUC are the field's own stated correct metrics, not something this project invented for convenience.
F-beta scoring for imbalanced problems	F2 (recall weighted twice as important as precision) formalizes exactly the recall-priority design philosophy this model's threshold-tuning already implements informally.
Classifier calibration: Platt scaling vs. isotonic regression (multiple ML references)	"Isotonic regression has been shown to work better than Platt scaling... when enough training data is available" — this project has ~78k training rows per horizon, well past that bar. Directly motivated §6.3/§8's calibration work.
SHAP for flood-model interpretability (2025 XAI literature)	"Addressing the black-box limitations that have hindered practical adoption by disaster management authorities" — SHAP explicitly framed as a trust requirement, not a nice-to-have.

3. Model Families Used in the Literature

Model family	Typical reported role	This project
Logistic Regression	Weakest baseline in most comparisons	Not used — literature consensus already rules it out for this kind of nonlinear, imbalanced problem.
Random Forest / XGBoost / LightGBM (tree ensembles)	Dominant choice across the field, "well-suited... due to gradient boosting on difficult-to-classify examples"	LightGBM used — matches the field's own stated reasoning for why.
LSTM / GRU / Transformer (deep sequence models)	Common for high-data-volume operational systems (Google Flood Hub: thousands of gauges)	Not used — this project has 30 stations, well below the data volume these architectures need to outperform gradient boosting.
Knowledge-guided hybrid (FHNN, inverse/forward catchment-state model)	One 2025 study, Microsoft-scale infrastructure	Not used — noted as a genuine future direction, not currently in scope.
LightGBM, per-horizon, pooled multi-station (this project)	—	24h: ROC-AUC 0.883, PR-AUC 0.214 — 72h: ROC-AUC 0.849, PR-AUC 0.261

4. Our Model: Design & Rationale

4.1 Three independent per-horizon models, pooled across all 30 stations

Rather than one sequence model spanning all three horizons, three separate binary classifiers are trained (24h/48h/72h) — the literature frames this as the "direct multi-step" strategy, now standard specifically because it avoids the error accumulation a recursive (sequence) approach suffers at longer horizons. All 30 stations are pooled into one training set per horizon (station/basin passed in as categorical features), not one model per station — independently matching a named best-practice rule from the LSTM streamflow literature: *"never train an LSTM on a single basin"* (Kratzert et al.) — several of the 30 stations have too few labeled rows to train an independent model each.

4.2 Handling rare, noisy labels

Positive labels are rare and come from multiple sources of differing confidence (GFMS satellite detections, DFO event catalogs, Global Flood Database). A custom sample-weighting scheme — computed from the "observed"-regime rows' real class balance, with an extra confidence discount for DFO-derived positives — replaces a blanket `class_weight="balanced"`, which the code's own analysis found would under-correct for how rare a positive actually is at deployment time.

4.3 Threshold tuned for recall, not accuracy

A missed flood costs far more than a false alarm, so each horizon's decision threshold is chosen to hit 85% recall rather than accepting whatever recall the default 0.5 threshold produces. Real result: at 24h, `threshold=0.534` yields `recall=0.850`, `precision=0.136` — a deliberate, documented trade of false alarms for caught floods, not an accident of an untuned threshold.

4.4 2026-08-14 additions: calibration and a naive-baseline comparison

Neither existed before this pass. **Isotonic calibration** (fit on a held-out validation slice, never the test set) recalibrates the *displayed* probability without touching the recall-tuned decision threshold above — those are different questions ("what decision to make" vs. "does the probability number mean what it says") and are kept deliberately separate. **Naive baselines** (climatology, persistence) were added because this model never had one, despite the project's own standing practice of comparing every model against a baseline (already applied to the discharge forecaster and the susceptibility model). Both are covered in full in §6 and §8.

5. Data Sources

Source	What it provides
Open-Meteo Forecast API (free, no key)	Live rainfall (local point + 9-point upstream basin grid) and soil moisture (0-7cm depth)
Open-Meteo Flood API (GloFAS reanalysis/forecast, free)	River discharge, local + upstream reference stations
GFMS (NASA), DFO, Global Flood Database, Copernicus GFM (SAR)	Historical flood-event labels — 4 independent sources, since no single one has continuous "confidently no flood" coverage across the full training period
MERIT Hydro (via Earth Engine, one-time sample)	Static per-station elevation/HAND terrain features

Train: 77,960 rows (17.2% positive, 24h horizon) with date < 2024-01-01. Test: 22,890 rows (4.3% positive) with date ≥ 2024-01-01 — a genuinely time-based split (never a random shuffle, which would leak nearby-day autocorrelation from test into train), restricted to the "observed"-regime label subset for a fair evaluation.

6. Results: Plots & Analytics

6.1 Discrimination: ROC, Precision-Recall, confusion matrix

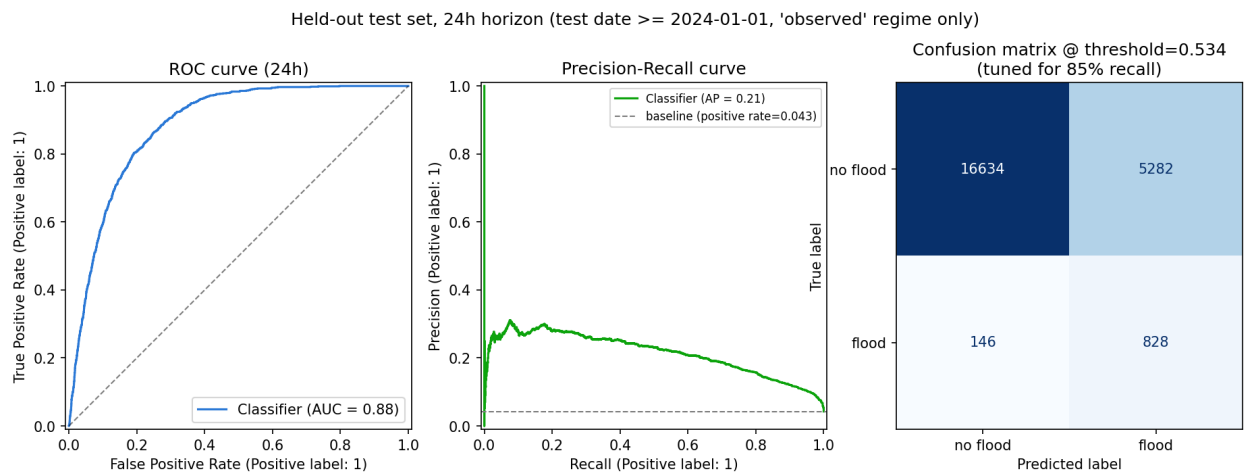


Figure — 24h horizon: ROC-AUC=0.883, PR-AUC=0.214, held-out test set (date ≥ 2024-01-01, 'observed' regime only).

Held-out test set, 48h horizon (test date $\geq$ 2024-01-01, 'observed' regime only)

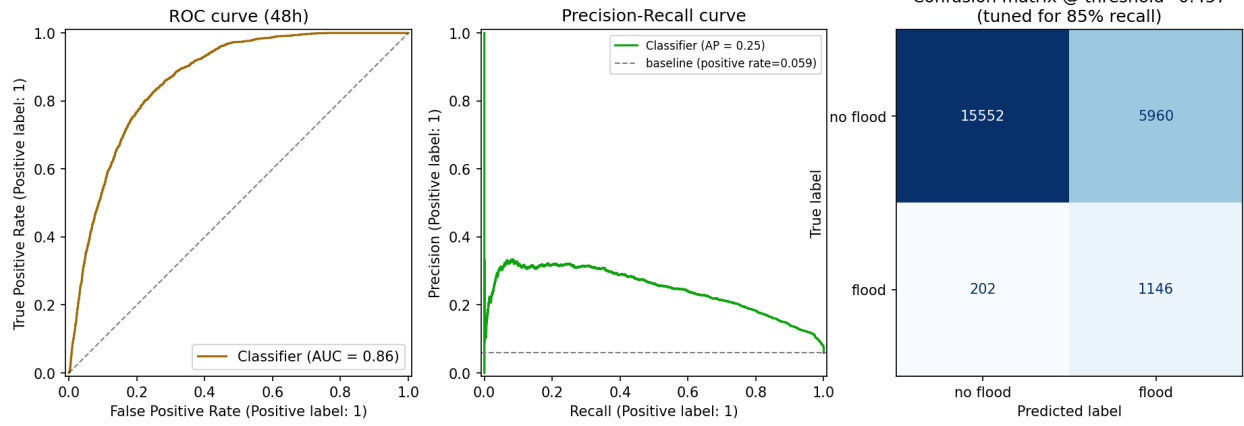


Figure — 48h horizon: ROC-AUC=0.864, PR-AUC=0.245, held-out test set (date $\geq$ 2024-01-01, 'observed' regime only).

Held-out test set, 72h horizon (test date $\geq$ 2024-01-01, 'observed' regime only)

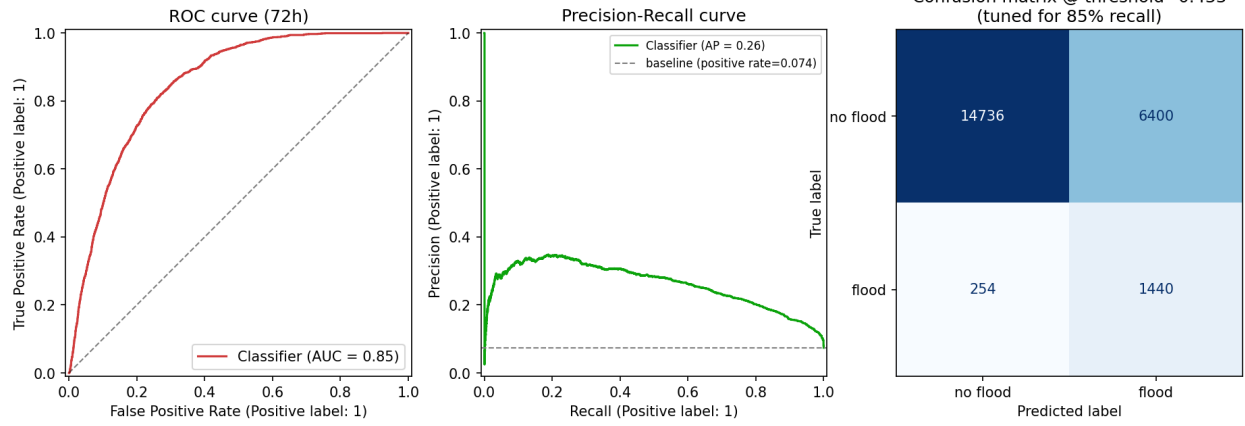


Figure — 72h horizon: ROC-AUC=0.849, PR-AUC=0.261, held-out test set (date $\geq$ 2024-01-01, 'observed' regime only).

6.2 Feature importance

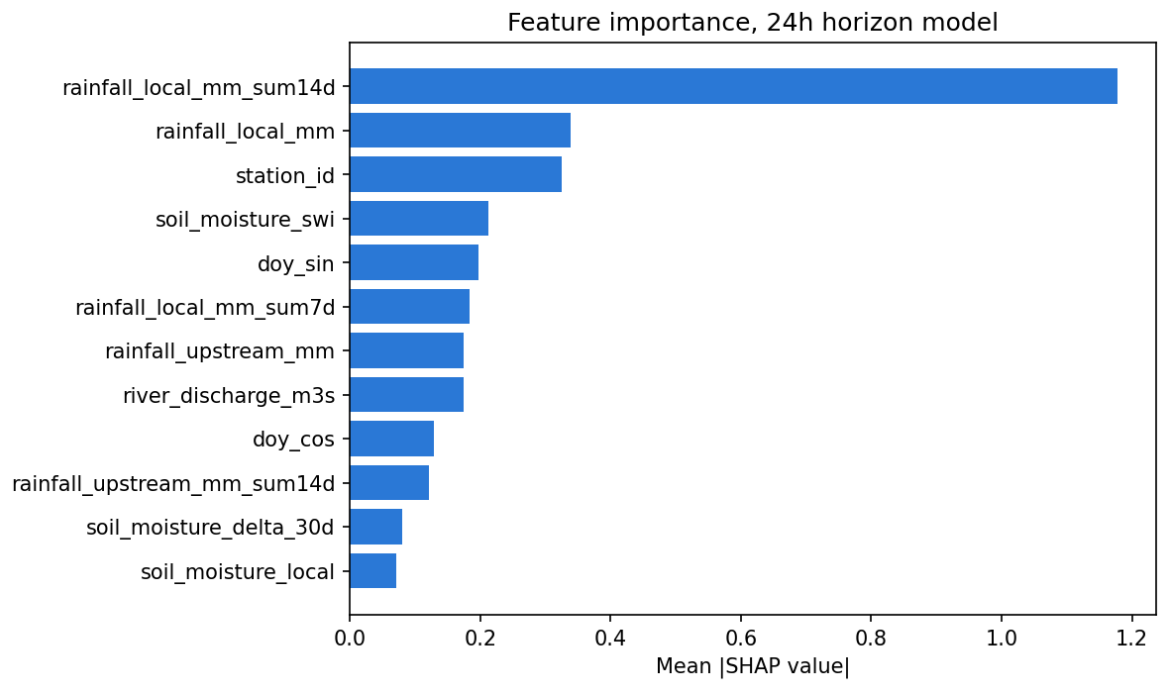


Figure — SHAP feature importance, 24h horizon model.

6.3 Calibration — deepened beyond a single Brier-score number

Three independent proper scoring rules were checked, not just one, so the "calibration helped" claim doesn't rest on a single metric that might happen to favor it:

Horizon	Brier (raw → calibrated)	Log loss (raw → calibrated)	ECE (raw → calibrated)
24h	0.1662 → 0.0385	0.4873 → 0.1373	0.2479 → 0.0164
48h	0.1671 → 0.0514	0.4863 → 0.1782	0.2361 → 0.0273
72h	0.1727 → 0.0660	0.4996 → 0.2547	0.2346 → 0.0394

All three metrics agree: calibration is a real, substantial improvement at every horizon — not an artifact of any one scoring rule.

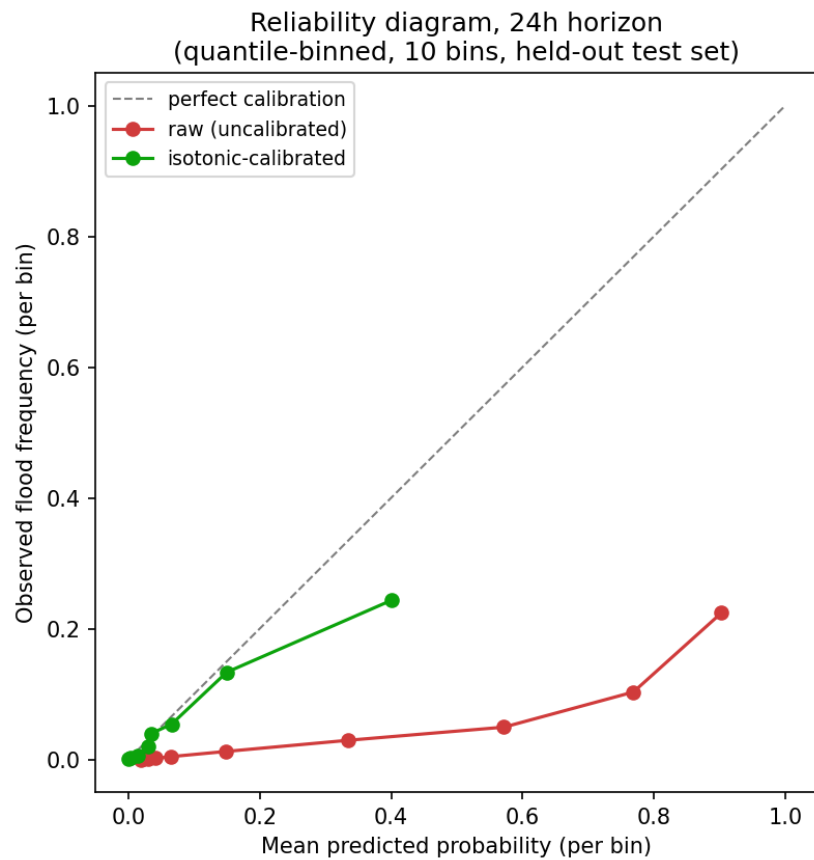


Figure — Reliability diagram, 24h horizon. The raw model is severely overconfident: a stated 90% probability corresponds to an observed flood frequency of only ~23%. Calibration closes most of that gap, though the calibrated curve still sits slightly below perfect (a bounded improvement, not a complete fix).

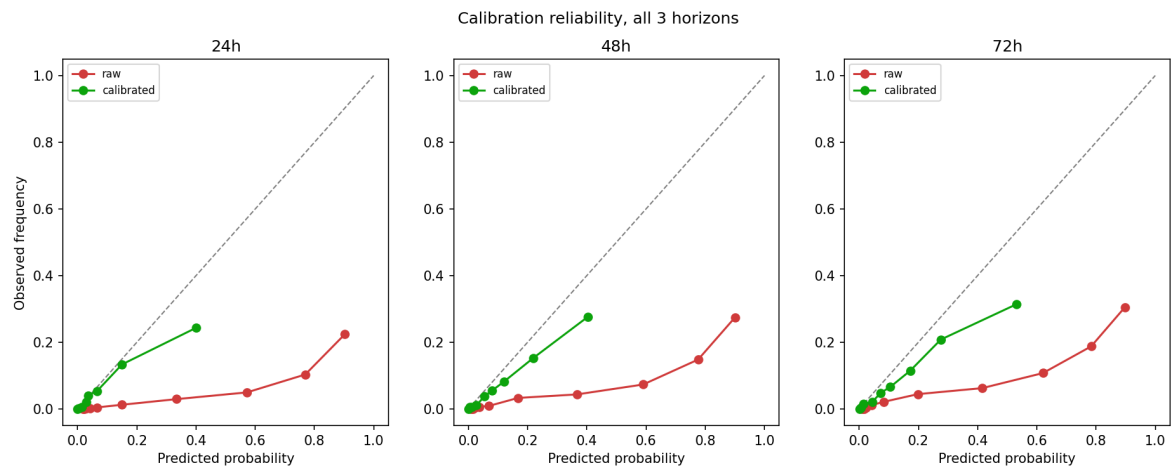


Figure — Reliability diagrams, all 3 horizons.

7. Comparison: This Model vs. the Literature

Several Bangladesh-specific studies (see the earlier flood-susceptibility report's §2.2 for the full list) report 97%+ accuracy for similar-sounding classification tasks. As documented in that report, most of those numbers come from a different, easier problem (static spatial classification, often randomly split) than this model's genuine temporal forecasting on a strict time-based split — the same honesty framing applies here and is not repeated in full.

Metric	This model (24h)	Literature context
ROC-AUC	0.883	0.85-0.98 range across reviewed studies, mostly without disclosed spatial/temporal-split discipline
PR-AUC	0.214	Rarely reported by comparable studies — most report only accuracy, which is why this model reports PR-AUC/recall instead (§2.1)
Precision @ 85% recall	0.136	Not directly comparable — reflects this model's own deliberate recall-priority design (§4.3), not a universal operating point

8. The Persistence-Baseline Finding

The most consequential result of the 2026-08-14 improvement pass, reported in full rather than minimized. Two naive, zero-live-data baselines were compared against the model on the exact same held-out test rows:

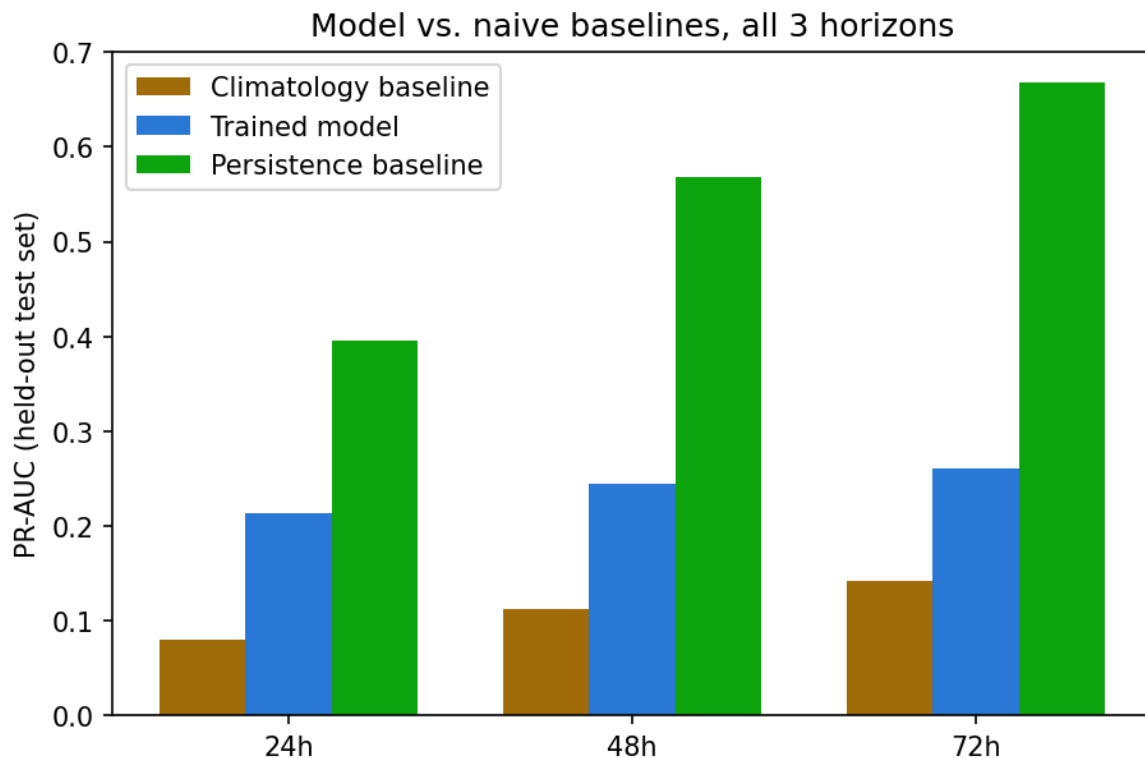


Figure — Model PR-AUC vs. climatology and persistence baselines, all 3 horizons.

The model clearly beats climatology (0.080 at 24h) at every horizon. It does **not** beat persistence (0.396 at 24h) — confirmed not an artifact of comparing mismatched operating points: at persistence's own natural 61.6% recall, the model's precision (20.7%) is still far below persistence's (61.6%).

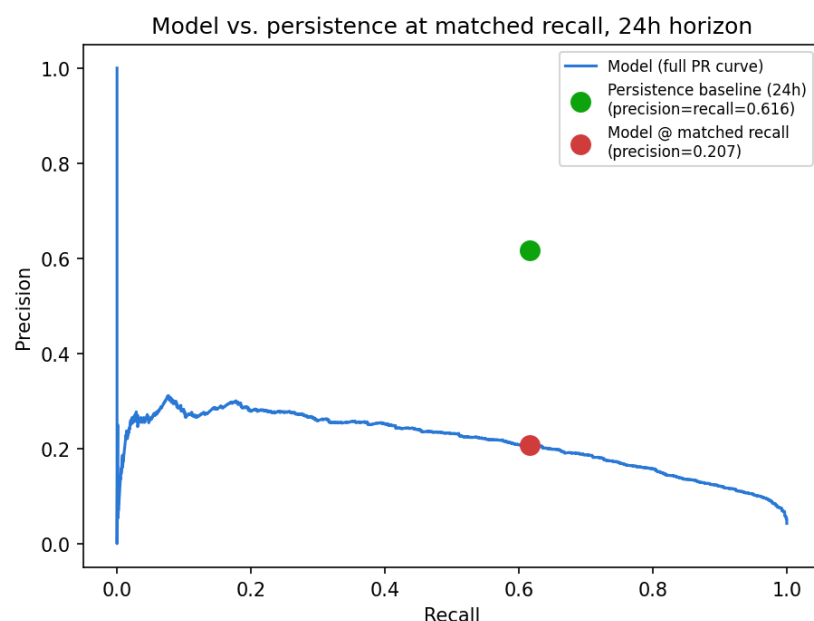


Figure — Model's full precision-recall curve vs. persistence at a matched recall point, 24h horizon.

Working explanation, not yet tested: flood_within_Nh labels cluster in multi-day contiguous blocks (real floods last many days), so persistence is unusually strong specifically at predicting the *continuation* of an

already-ongoing flood — exactly the part of the problem requiring the least forecasting skill. The model's real value is plausibly concentrated in predicting flood *onset* (a brand-new event persistence can never predict, since it only ever repeats yesterday's label). A pooled PR-AUC across both continuation and onset rows lets the easier continuation majority dominate the metric and mask genuine onset-specific skill. This has not yet been tested — it is the clear next investigation for this model, not something to quietly work around.

9. Limitations & Honesty Notes

- **Does not beat a persistence baseline on pooled PR-AUC (§8)** — the single most important limitation in this report, not buried in a list.
- **Calibration is bounded, not perfect**: even after isotonic calibration, the model rarely produces very high probabilities (the calibrated reliability curve tops out around 40% mean predicted probability at 24h) — a real improvement, not a complete fix.
- **Calibrator is not yet wired into live serving** — saved as an artifact per horizon (e.g. `model_24h_calibrator.joblib`) but the live API still returns the raw score.
- **Precision at the tuned threshold is genuinely low** (13-18% across horizons) — a disclosed, structural consequence of rare positive labels and a deliberate recall-priority design, not a bug.

10. Future Work

- **Highest priority**: design and test an onset-vs-continuation-aware evaluation (and possibly training target) directly motivated by §8.
- Wire the saved isotonic calibrators into live serving so the API returns a calibrated, not raw, probability.
- The cross-correlation lag experiment (tested 2026-08-14, rejected — extending [1,2,3,5] with lags 7 and 12 was neutral-to-harmful) closes that avenue; no further lag-design work planned unless new evidence emerges.

Appendix: Reproducibility Reference

Artifact	Path
Feature engineering	backend/train/build_features.py
Model training + evaluation	backend/train/train_model.py
Cross-correlation lag experiment (rejected)	backend/train/experiment_cross_correlation_lags.py
Trained models + metrics + calibrators	backend/models/2026-08-07c/
Standalone service	packages/flood-risk-classifier/
This report's figures	reports/generate_classifier_assets.py
Full build log	MODEL_BUILD_PLAN.md (2026-08-14 entries)